AN IDEMPOTENT ON THE FIBRE OF A PROPER MORPHISM COMES FROM THE PUSHFORWARD OF THE STRUCTURE SHEAF

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Allowed standard prerequisites (editorial).

- The formal functions theorem identifies the completion of the stalk of the pushforward of the structure sheaf with the inverse limit of the sections of the infinitesimal neighbourhoods of the fibre. invoked at proof lines 8341-8348. Produces the compatible system of idempotents whose limit is the required element of the stalk.
- Sections and idempotents can be descended along a proper limit of schemes of finite presentation over a Noetherian base. invoked at proof lines 8380-8390. Reduces the general case to the Noetherian case.

Source reference. Original lemma 02OD: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 8345-8349 to identify the completed stalk with the inverse limit of the sections of the infinitesimal neighbourhoods; the construction of the compatible idempotents is inside the counted span.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 8361-8365 to lift the idempotent of the completed stalk to the stalk itself; the verification that its image is the original idempotent is inside the counted span.

Source reference. Original lemma 0A0P: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8378-8384 to reduce the general case to the Noetherian case; the reduction itself is inside the counted span.

Source reference. Original lemma 01Z0: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 8385-8388 to descend the section produced in the Noetherian case; the conclusion drawn for the general case is inside the counted span.

Source reference. Original lemma 0CUG: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer within the limit reduction; the reduction and the final conclusion are inside the counted span.

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Source reference. Original lemma 002M: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer for the limit manipulation; the actual reduction to the Noetherian case is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a proper morphism of schemes. Let $s \in S$ and let $e \in H^0(X_s, \mathcal{O}_{X_s})$ be an idempotent. Then e is in the image of the map $(f_*\mathcal{O}_X)_s \to H^0(X_s, \mathcal{O}_{X_s})$.*

Proof. Let $X_s = T_1 \amalg T_2$ be the disjoint union decomposition with T_1 and T_2 nonempty and open and closed in X_s corresponding to e, i.e., such that e is identitically 1 on T_1 and identically 0 on T_2.

Assume S is Noetherian. We will use the theorem on formal functions in the form of Cohomology of Schemes, Lemma 02OD. It tells us that

$$(f_*\mathcal{O}_X)^\wedge_s = \lim_n H^0(X_n, \mathcal{O}_{X_n})$$

where X_n is the nth infinitesimal neighbourhood of X_s. Since the underlying topological space of X_n is equal to that of X_s we obtain for all n a disjoint union decomposition of schemes $X_n = T_{1,n} \amalg T_{2,n}$ where the underlying topological space of $T_{i,n}$ is T_i for $i = 1, 2$. This means $H^0(X_n, \mathcal{O}_{X_n})$ contains a nontrivial idempotent e_n, namely the function which is identically 1 on $T_{1,n}$ and identically 0 on $T_{2,n}$. It is clear that e_{n+1} restricts to e_n on X_n. Hence $e_\infty = \lim e_n$ is a nontrivial idempotent of the limit. Thus e_∞ is an element of the completion of $(f_*\mathcal{O}_X)_s$ mapping to e in $H^0(X_s, \mathcal{O}_{X_s})$. Since the map $(f_*\mathcal{O}_X)^\wedge_s \to H^0(X_s, \mathcal{O}_{X_s})$ factors through $(f_*\mathcal{O}_X)^\wedge_s/\mathfrak{m}_s(f_*\mathcal{O}_X)^\wedge_s = (f_*\mathcal{O}_X)_s/\mathfrak{m}_s(f_*\mathcal{O}_X)_s$ (Algebra, Lemma 05GG) we conclude that e is in the image of the map $(f_*\mathcal{O}_X)_s \to H^0(X_s, \mathcal{O}_{X_s})$ as desired.

General case: we reduce the general case to the Noetherian case by limit arguments. We urge the reader to skip the proof. We may replace S by an affine open neighbourhood of s. Thus we may and do assume that S is affine. By Limits, Lemma 0A0P we can write $(f : X \to S) = \lim(f_i : X_i \to S_i)$ with f_i proper and S_i Noetherian. Denote $s_i \in S_i$ the image of s. Then $s = \lim s_i$, see Limits, Lemma 0CUG. Then $X_s = X \times_S s = \lim X_i \times_{S_i} s_i = \lim X_{i,s_i}$ because limits commute with limits (Categories, Lemma 002M). Hence e is the image of some idempotent $e_i \in H^0(X_{i,s_i}, \mathcal{O}_{X_{i,s_i}})$ by Limits, Lemma 01Z0. By the Noetherian case there is an element $\tilde{e}_i$ in the stalk $(f_{i,*}\mathcal{O}_{X_i})_{s_i}$ mapping to e_i. Taking the pullback of $\tilde{e}_i$ we get an element $\tilde{e}$ of $(f_*\mathcal{O}_X)_s$ mapping to e and the proof is complete. □